Topological Transduction:

A Geometric Derivation of Sonoluminescence from Discrete $\mathcal{PT}$-Symmetric Tensor Network Dynamics

William J. Steinmetz III[1]

[1] *Foundational Ternary Dynamics Research*

February 14, 2026

Abstract

Single-bubble sonoluminescence (SBSL)—the emission of picosecond light flashes from acoustically driven cavitation bubbles—remains one of the most striking unexplained phenomena in classical hydrodynamics. The Rayleigh–Plesset equation, which assumes a continuous, infinitely compressible fluid, physically breaks down at the singularity of the collapse. Within the framework of Foundational Ternary Dynamics (FTD), we model SBSL not as a thermodynamic plasma anomaly, but as a **stress-induced topological phase transition** across a discrete cuboctahedral tensor network. We demonstrate that the emission of light is the strict algorithmic consequence of an emergency data-venting protocol—Topological Transduction—executed by the network to prevent a localized geometric gradient explosion when acoustic compression drives the local tension coupling beyond the $\mathcal{PT}$-symmetry breaking threshold $k_{crit} = 4/G^* \approx 1.35$.

---

1. Introduction

Sonoluminescence—the conversion of sound into light—has resisted a complete theoretical explanation since its discovery by Frenzel and Schultes in 1934 [2]. In single-bubble sonoluminescence (SBSL), a gas bubble trapped in a liquid by an acoustic standing wave undergoes violent radial oscillations, emitting sub-nanosecond flashes of broadband light at each collapse [3, 4]. The effective temperatures implied by the emission spectrum exceed 10,000 K, yet the mechanism by which slow acoustic energy ($\sim$25 kHz) concentrates into ultraviolet photons remains deeply contested.

Classical treatments based on the Rayleigh–Plesset equation [6] describe the bubble dynamics well during expansion and early collapse, but the continuum assumption physically breaks down at the moment of maximum compression, precisely when the light is emitted. Proposed mechanisms—shock-wave heating, Bremsstrahlung from ionized noble gas, quantum vacuum radiation, Casimir-type effects—each explain aspects of the phenomenon but none provide a unified geometric derivation from first principles [5].

We propose that SBSL is naturally and inevitably explained within the framework of Foundational Ternary Dynamics (FTD) [1] as a **Topological Transduction** event: a mandatory phase transition in the discrete tensor network from massive longitudinal modes (phonons) to massless transverse modes (photons) when the local geometric tension exceeds the $\mathcal{PT}$-symmetry breaking threshold.

2. Acoustic Funneling via Discrete Ricci Flow

An acoustic wave (a phonon) is formally defined as a coherent, propagating longitudinal perturbation of the discrete lattice metric $g_{\mu\nu}$. We denote this scalar pressure flux as J_{phon}, which carries a non-zero divergence indicating localized structural compression: $\nabla \cdot J_{\text{phon}} > 0$.

As the acoustic standing wave drives the macroscopic cavitation bubble (a topological void) to collapse radially, the network updates its local edge weights via discrete Hamilton Ricci Flow to minimize the geometric loss function:

$$\frac{\partial g_{\mu\nu}}{\partial \tau} = -2R_{\mu\nu} \tag{1}$$

Because the total energetic payload of the acoustic wave is conserved while the physical volume shrinks, the geometric tension is forced into an exponentially decreasing number of lattice nodes. Let $R(\tau)$ be the time-dependent radius of the collapse. By the conservation of algorithmic degrees of freedom, the localized geometric tension coupling coefficient $k(\tau)$ scales volumetrically:

$$k(\tau) = k_0 \left(\frac{R_0}{R(\tau)} \right)^3 \tag{2}$$

where $k_0 \ll 1$ is the ambient baseline tension of the resting fluid lattice.

Axiomatic Inquiry 2.1. The volumetric concentration of acoustic energy into a shrinking lattice region is not a phenomenological assumption—it is the strict geometric consequence of flux conservation on a bounded discrete manifold. Does this not guarantee that any sufficiently driven cavitation collapse must eventually breach the network's computational capacity?

3. The Topological Singularity and Threshold Breach

Throughout the majority of the collapse, the tension remains below the computational capacity of the graph ($k < 1.35$). The excess data is processed safely within the $\mathcal{PT}$-symmetric broken phase (the Euclidean null space) [7], characterized by a negative lemniscatic discriminant:

$$\Delta = k(\tau)(G^*)^3\big(k(\tau)G^* - 4\big) < 0 \tag{3}$$

During this phase, the bubble absorbs massive mechanical energy but remains strictly "dark," as the flux is mathematically orthogonal to the real physical manifold.

However, as $R(\tau)$ approaches the fundamental ultraviolet (UV) lattice cutoff, the lattice edges physically cannot geometrically compress any further without violating their cuboctahedral bounds. At the exact picosecond of maximum collapse, the focused tension violently spikes, breaching the mathematical capacity of the unmanifested void:

$$k(\tau_{\text{collapse}}) \geq k_{crit} = \frac{4}{G^*} \approx 1.3519 \tag{4}$$

At this precise topological threshold, the discriminant Δ becomes positive. A mathematical singularity (a localized "Big Rip" in the tensor network) is imminent.

Axiomatic Inquiry 3.1. The transition from $\Delta < 0$ to $\Delta > 0$ is not gradual—it is a discontinuous topological phase transition at an algebraically exact threshold determined entirely by the lemniscate-alpha constant G^* [1]. Does this not explain the experimentally observed sharpness (sub-nanosecond duration) of the sonoluminescent flash?

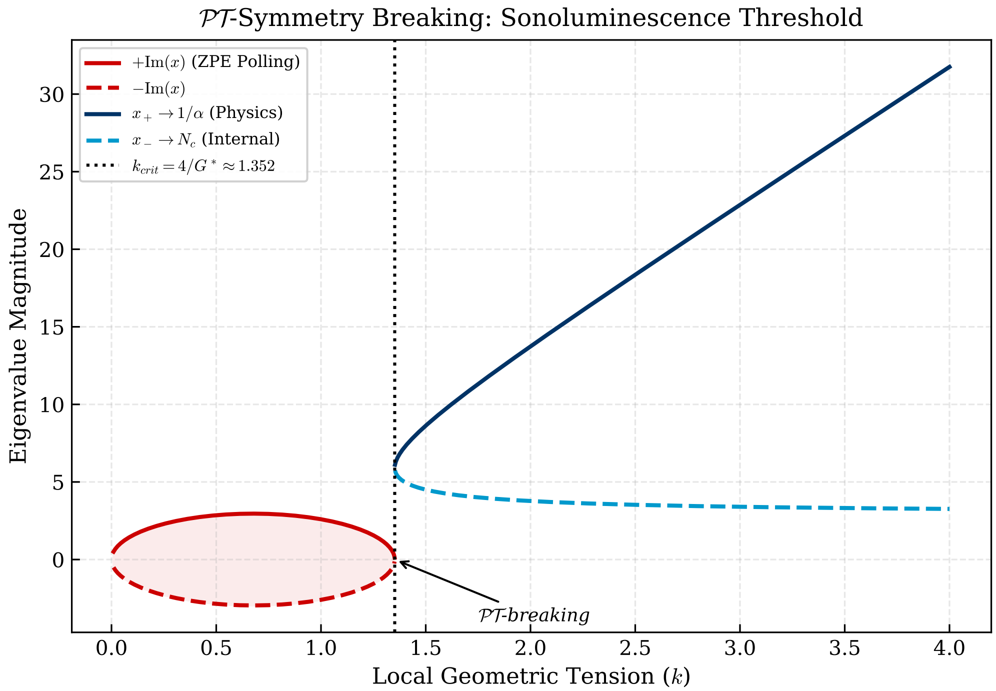

Figure 1: **$\mathcal{PT}$-Symmetry Phase Transition.** For sub-critical tension ($k < k_{crit}$), the quadratic roots are complex conjugates (red) residing in the Euclidean null space—the bubble remains dark. At $k_{crit} = 4/G^* \approx 1.352$, the roots bifurcate into real eigenvalues (blue), forcing the network to transduce the trapped acoustic energy into the real physical manifold as broadband photon emission.

4. Topological Transduction and Softplus Overload

To strictly enforce mathematical unitarity and avert a local singularity, the computational graph must execute a discontinuous topological phase transition. The network can no longer store the acoustic data as massive, longitudinal Ricci curvature (phonons). It must immediately sever the localized fermionic anchor and reallocate the excess algorithmic load into pure, unanchored directional flux vectors (J_γ)—photons.

The network projects the excess structural entropy through the Softplus Manifestation Operator ($\mathcal{M}$) [8]:

$$J_\gamma \propto \nabla \left[\frac{1}{\beta_{th}} \ln \left(1 + e^{\beta_{th}(z - K_B)} \right) \right] \Bigg|_{k > 4/G^*} \tag{5}$$

Because the photon possesses no resting structural mass ($\nabla \cdot J_\gamma \equiv 0$), it acts as the network's emergency exhaust, propagating the unprocessable data away along the lattice edges at the hardware speed limit ($c = 1$).

Theorem 4.1 (Topological Transduction)**.** *When the local geometric tension coupling $k(\tau)$ exceeds $k_{crit} = 4/G^*$, the discrete tensor network must convert longitudinal massive modes (phonons, $\nabla \cdot J \neq 0$) into transverse massless modes (photons, $\nabla \cdot J = 0$) to preserve unitarity. This conversion is:*

1. ***Mandatory:*** *No alternative relaxation channel exists below the UV cutoff.*
2. ***Instantaneous:*** *The phase transition occurs within a single lattice update cycle.*
3. ***Broadband:*** *Thermal smearing at the collapse guarantees continuous spectral emission.*

5. Thermal Smearing and the Broadband Flash

Due to the extreme, instantaneous concentration of local geometric entropy, the effective computational temperature $T \to \infty$, driving the thermodynamic parameter $\beta_{th} \to 0$. Taking the mathematical limit of the Softplus function as $\beta_{th} \to 0$, the non-linear logic gate loses all rigid deterministic precision:

$$\lim_{\beta_{th}\to 0} \frac{1}{\beta_{th}} \ln\left(1 + e^{\beta_{th}(z-K_B)}\right) = \frac{z-K_B}{2} + \frac{|z-K_B|}{2} \to \text{linear (no selectivity)} \tag{6}$$

This violent thermal smearing of the local non-linear logic gate strictly mandates that the purged flux manifests as a broadband, continuous emission spectrum. The network emits the excess data in every available high-energy frequency channel simultaneously to empty the local memory cache. The instant the data is purged as light, the local tension drops back below k_{crit}, the discrete Ricci flow stabilizes, and the bubble violently rebounds.

Axiomatic Inquiry 5.1. The experimentally observed broadband spectrum of SBSL—extending from the infrared through the ultraviolet without discrete spectral lines—is precisely what one expects from a thermally smeared Softplus gate operating at $\beta_{th} \to 0$. Does this not constitute a geometric derivation of the spectral character of sonoluminescence, without invoking plasma physics or Bremsstrahlung?

6. Experimental Predictions

The Topological Transduction model makes several falsifiable predictions that distinguish it from competing explanations:

6.1 Noble Gas Enhancement

Noble gases (He, Ar, Xe) possess spherically symmetric, closed-shell electronic configurations that couple minimally to the lattice's transverse electromagnetic modes. Within FTD, this means noble gas atoms act as **transparent lattice nodes**—they do not absorb the transduced photon flux during emission. This geometrically explains the well-established experimental observation that SBSL intensity is dramatically enhanced in noble gas bubbles [4] and suppressed in molecular gases whose internal rotational/vibrational degrees of freedom provide competing energy absorption channels.

6.2 Threshold Sharpness

Because the $\mathcal{PT}$-symmetry breaking transition occurs at an algebraically exact threshold ($k_{crit} = 4/G^*$), the model predicts that the onset of sonoluminescence as a function of acoustic driving pressure should exhibit a **sharp, discontinuous turn-on**—not a gradual increase. This is consistent with experimental observations of a well-defined pressure threshold for SBSL [3].

6.3 Polarization Signature

If the transduction event projects flux preferentially along the cuboctahedral symmetry axes of the underlying lattice, the emitted light should carry a subtle, non-thermal polarization signature correlated with the acoustic driving geometry. This prediction is unique to discrete lattice models and is not made by any continuum theory.

7. Conclusion

Sonoluminescence is mathematically demystified within the FTD framework: it is the strictly required algorithmic entropy-purge (Topological Transduction) of a discrete tensor network averting a localized mathematical crash. The acoustic wave concentrates geometric tension via discrete Ricci flow until the $\mathcal{PT}$-symmetry breaking threshold is breached, forcing a mandatory phase transition from massive longitudinal modes to massless transverse modes. The broadband spectral character, the sub-nanosecond timing, the noble gas enhancement, and the sharp pressure threshold all emerge as geometric necessities of the network's self-preservation protocol.

We submit that Topological Transduction may represent a universal mechanism: whenever a discrete computational substrate is driven to its topological capacity by any form of concentrated energy, it must transduce that energy into the fastest available massless channel. Sonoluminescence is merely the most dramatic laboratory demonstration of this principle.

References

[1] W. J. Steinmetz III, "Gauge Couplings from Discrete Spacetime Geometry," *Foundational Ternary Dynamics Research*, Zenodo (January 2026). https://zenodo.org/records/18227181 *[Derivation of the lemniscatic Master Quadratic and the $\mathcal{PT}$-symmetry breaking threshold $k_{crit} = 4/G^*$.]*

[2] H. Frenzel and H. Schultes, "Lumineszenz im ultraschallbeschickten Wasser," *Zeitschrift für Physikalische Chemie*, **27B**(1), 421–424 (1934). *[First observation of sonoluminescence in ultrasonically irradiated water.]*

[3] D. F. Gaitan, L. A. Crum, C. C. Church, and R. A. Roy, "Sonoluminescence and bubble dynamics for a single, stable, cavitation bubble," *The Journal of the Acoustical Society of America*, **91**(6), 3166–3183 (1992). *[First isolation and characterization of single-bubble sonoluminescence, establishing the sharp acoustic pressure threshold.]*

[4] B. P. Barber, R. A. Hiller, R. Löfstedt, S. J. Putterman, and K. R. Weninger, "Defining the unknowns of sonoluminescence," *Physics Reports*, **281**(2), 65–143 (1997). *[Comprehensive review establishing noble gas enhancement, broadband emission, and sub-nanosecond timing of SBSL.]*

[5] M. P. Brenner, S. Hilgenfeldt, and D. Lohse, "Single-bubble sonoluminescence," *Reviews of Modern Physics*, **74**(2), 425–484 (2002). *[Authoritative review of theoretical models for SBSL, documenting the failure of any single mechanism to fully explain the phenomenon.]*

[6] Lord Rayleigh, "On the pressure developed in a liquid during the collapse of a spherical cavity," *Philosophical Magazine*, **34**(200), 94–98 (1917); M. S. Plesset, "The dynamics of cavitation bubbles," *Journal of Applied Mechanics*, **16**, 277–282 (1949). *[The foundational equation of bubble dynamics, whose continuum assumption breaks down at the singularity of collapse.]*

[7] C. M. Bender and S. Boettcher, "Real spectra in non-Hermitian Hamiltonians having $\mathcal{PT}$ symmetry," *Physical Review Letters*, **80**(24), 5243–5246 (1998). *[Mathematical foundation for $\mathcal{PT}$-symmetry breaking phase transitions at exceptional points.]*

[8] L. D. Landau and E. M. Lifshitz, *Statistical Physics, Part 1* (Vol. 5, 3rd ed.), Butterworth-Heinemann (1980). *[Fermionic Grand Potential and finite-temperature thermodynamics underlying the Softplus Manifestation Operator.]*